Supplementary material: Register-aware computational prioritization of EBV–self peptide pairs across HLA class II contexts

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1 Supplementary Methods

1.1 SM1. Frozen-core input and nominally TCR-exposed BLOSUM62 score

The initial 6,400-comparison V3 matrix stores one declared P1–P9 core for each peptide arm. These cores were frozen before the score matrix was ranked and were not replaced by the later audit-predictor calls. The initial screen is therefore conditional on the declared-core input. The local V2 register-prediction records identify the caller as `IEDB recommended_binding` and retain the allele, peptide, returned `core_peptide`, percentile rank, IC50, core-start resolution, and raw-response filename. The IEDB service's internal model version, exact invocation date, and service-side tie behavior were not retained; the frozen core input is therefore auditable from the saved response records but not fully rerunnable against an identically versioned remote service.

For two declared nine-residue cores a and b, the nominally TCR-exposed normalized BLOSUM62 score was calculated at $I = \{\text{P2}, \text{P3}, \text{P5}, \text{P7}, \text{P8}\}$ as $S(a,b) = \sum_{i \in I} B(a_i, b_i) \,/\, \sum_{i \in I} \max\{B(a_i, a_i), B(b_i, b_i)\}$. B denotes the published BLOSUM62 substitution value. The numerator sums the five matched residue-pair values; the denominator sums the larger of the two residue-specific self-substitution values at each position. Exact identities are handled by the same formula, which gives the upper bound of 1.0; conservative and nonconservative substitutions lower the numerator relative to that bound. Cores containing nonstandard or ambiguous amino-acid symbols were excluded during curation. The score is a relative, register-conditional prioritization quantity, not a probability of presentation, recognition, cross-reactivity, or molecular mimicry.

The V3 ranking was lexicographic: nominally TCR-exposed BLOSUM62 descending, physicochemical mismatch ascending, nominally TCR-exposed identity descending, local-surface percentile ascending, and lexical pair identifier ascending. Physicochemical mismatch was a scaled comparison of charge, hydropathy, donor, acceptor, and aromaticity at the five nominally exposed positions. Local-surface percentile was not an independent assay or a global structural score. After Kabsch alignment of the first 85 Cα atoms from each HLA alpha and beta chain, six local peptide-surface features were calculated: side-chain solvent-exposure mismatch, exposure-weighted centroid RMSD, exposure-weighted side-chain orientation RMSD, exposed-residue distance-matrix RMSD, exposure-weighted chemistry mismatch, and exposure-weighted backbone RMSD. Glycine used Cα as a side-chain fallback. Solvent accessibility used a deterministic 32-point Shrake–Rupley approximation with a 1.4 Å probe. Each feature was summarized at its model-combination 75th percentile, converted to a within-allele percentile, and the six percentiles were averaged. This quantity could break a prior sequence tie only when the model was complete and passed the declared fixed-coordinate register-sensitivity rule; it was otherwise blank. Binding-prediction ranks, full-core RMSD, and anchor RMSD were diagnostics and did not enter the V3 rank.

An ablation audit of the 6,400-pair V3 table quantified the role of each tie key. Within the four allele-specific 1,600-pair cohorts, 6,117 candidates were in a primary-BLOSUM tie. Physicochemical mismatch resolved 6,055 of these candidates; nominally exposed identity then resolved 25 more, and the local-surface percentile resolved

a further 166. Thus, the modeled surface descriptor affected ordering only after three sequence-based keys and did not define the primary score; lexical pair identifier resolved any residual exact ties. This is a deterministic ordering device, not an effect-size model or a claim that structural annotations are reliable enough to validate peptide registers.

1.2 SM2. Enriched-panel construction and provisional-stratum rules

1.2.1 Enriched 49-pair evaluation panel

The enriched 49-pair evaluation panel was analyzed separately from the initial eight-pair evidence audit and is not an independent validation cohort. All final pair identifiers matched entries in the original 6,400-comparison V3 matrix. The frozen source pool comprised eight Stage-1 evidence-audit candidates, ten previously retained near-miss candidates, and 31 rank-based prefilter candidates from the V3 matrix. Deduplication by allele, declared viral core, and declared self core reduced these 49 source rows to 47 unique pair definitions. Two deterministic replacements were then selected from the relevant within-allele V3 rankings to restore the fixed 49-pair panel; each replacement had pre-existing viral- and self-arm binding percentile ranks $\leq 20\%$. Consequently, later binding-gate pass rates are conditioned on this enrichment and must not be interpreted as estimates for the original 6,400-pair universe. The final panel contained seven HLA-DRB1*03:01 pairs, 14 HLA-DRB1*08:01 pairs, 16 HLA-DRB1*13:03 pairs, and 12 HLA-DRB1*15:01 pairs, corresponding to 98 peptide arms.

1.2.2 Secondary provisional-tier rules

Secondary provisional strata were assigned after the enriched 49-pair panel as routing labels for subsequent experimental binding and register-mapping work. A strict peptide arm required all three selected computational-output percentile ranks to be $\leq 20\%$, agreement of at least two outputs on the same nine-residue core, and agreement between that core and the declared core. A near-passing arm required declared-core agreement and at least two of three output percentile ranks $\leq 20\%$. These are operational quantiles, not tool-calibrated affinity thresholds, and output percentiles are not assumed to have identical biological meaning across tools. Computational stratum A (formerly "strong-medium") required strict support for both arms plus exact-sequence, exact-allele direct-binding evidence for at least one arm. Computational stratum B (formerly "weak-medium") required either strict support for both arms without verified exact-allele binding or ligand evidence, or support for both declared registers across two calibrated same-allele PANDORA templates and expected/adjacent-register decoys despite one narrow predictor discrepancy. Computational stratum C (formerly "strong-low") required strict support for one arm and near-passing support for the other. Candidates failing these conditions were left untiered. No candidate received high-priority status because the automated public-evidence endpoint was unavailable under the frozen review protocol. These strata do not establish peptide-HLA affinity, natural presentation, T-cell recognition, cross-reactivity, molecular mimicry, or disease relevance.

1.3 SM3. Statistical calibration and sensitivity analyses

Statistical methods were implemented to test whether observed nominally TCR-exposed BLOSUM62 scores exceeded scores expected after disruption of residue-position alignments. Each HLA allele had its own separate simulation with a starting universe of 40 EBV cores and 40 self cores, making 1,600 pairwise comparisons. Ten thousand simulation replicates were completed independently for each allele at random seed `20260914`. These peptides' positions were classified as HLA anchor positions (P1, P4, P6, P9) or nominally TCR-exposed/core non-anchor positions (P2, P3, P5, P7, P8). When making the replicates, HLA anchor positions were held in place while the five nominally exposed residues were independently permuted within each 9-mer core and each peptide's exposed-residue composition was unchanged by this within-core permutation. Furthermore, each unique peptide

was shuffled once per replicate and was then reused across 40 pairings to preserve the Cartesian dependence structure. Then, a recalculation was performed with the unchanged normalized BLOSUM62 score function, with higher scores being ranked higher, and five prespecified outputs were collected: pair-specific upper-tail Monte Carlo p, Benjamini-Hochberg q across eight audited pairs, within-allele maxT family-wise p, the null median, and the 95% null interval. Random re-pairing was excluded because the observed 40×40 matrix already contained every possible pairing.

Next, presentation-support threshold sensitivity of the frozen 49-pair panel was evaluated using IEDB recommended-binding output, NetMHCIIpan 4.3 EL output, and MixMHC2pred 2.1 context output. NetMHCIIpan and MixMHC2pred versions and targets are retained in the raw response records; the IEDB service call retains its output but not an internal model version, exact access date, or service-side tie behavior. The IEDB call is therefore an archival dependency rather than a fully rerunnable third model. The specific operational percentile thresholds evaluated were 1%, 2%, 5%, 10%, and 20%; they were not calibrated as common affinity thresholds across tools. The routing requirements were all three selected outputs passing for both peptide arms or at least two of three outputs passing for both arms. Alongside these rules, two register criteria were evaluated as separate grid conditions: each output's predicted core start was required either to be identical to the declared core start or to have an absolute start-position difference of ≤ 1 residue. Outputs were treated as a correlated computational ensemble, not as independent biological measurements.

The frozen grid was rerun from all 490 continuous predictor records (49 pairs $\times$ two arms $\times$ five recorded outputs). Under the locked all-three-output, both-arm rule, 0/49 pairs passed at 1%, 2%, 5%, or 10%, and 2/49 passed at the explicitly permissive 20% cutoff; the two passing pairs were BALF5–TALDO1. Under the two-of-three-output, both-arm rule, the DRB1*15:01 pair passed at 10%, whereas the DRB1*13:03 pair first passed at 20%. The ± 1 register rule added no candidates to the strict all-three-output condition; it added one nonlead candidate only in the relaxed two-of-three 20% condition. The threshold is a routing convention for a deliberately enriched panel, not a predicted-affinity category, and the three outputs were not treated as independent evidence.

Four-metric rank sensitivity was also performed on the four allele-specific cohorts containing 1,600 pairs each. Pairs were evaluated using nominally TCR-exposed BLOSUM62, full-core BLOSUM62, nominally TCR-exposed sequence identity, and Grantham distance. Higher similarity was better for BLOSUM62 and identity, while lower Grantham distances indicated greater physicochemical similarity. Standard Grantham distance [9] was used as the physicochemical metric. The pairwise score was calculated by averaging published positional distances between matched nominally exposed residues at P2, P3, P5, P7, and P8. Existing calibration covered eight panels from three independent TCR systems and did not include verified specificity negatives; it supports only a provisional physicochemical sensitivity comparison, not validation of the primary score as a cross-reactivity predictor. A pair was multi-metric rank-concordant if it was within the top 1% under at least three metrics and partially rank-concordant if it did not meet that criterion but was within the top 5% under at least three metrics. For tie handling, average ranks were used to calculate pairwise within-allele Spearman correlations when ties occurred, and competition ranks were used for these labels. Consequently, rank ≤ 16 can include more than 16 pairs when scores tie; the V3 matrix retains the actual rank and tie size. The top 5% boundary was rank ≤ 80.

The constrained null answers a narrow conditional question: whether the observed five-position alignments are unusually favorable after within-core exposed-position permutation in this curated universe. Each core has at most $5! = 120$ unique exposed-residue permutations; 10,000 replicates therefore sample this finite set repeatedly while preserving the matrix's Cartesian dependence structure. The procedure is not a proteome-wide null, a matched-decoy test, a test of natural presentation, or a test of TCR cross-reactivity.

1.4 SM8. Calibration and exploratory modeling

The literature-derived Hy.2E11 cross-reactive benchmark, $\text{BALF5}_{627\text{-}641}$-$\text{MBP}_{85\text{-}99}$, served as a structural-comparison reference [11]. The benchmark contains HLA-DRA*01:01-HLA-DRB5*01:01 for BALF5 (PDB:

1H15) and HLA-DRA*01:01-HLA-DRB1*15:01 for MBP (PDB: 1BX2). It was therefore treated as a DR15 haplotype-level control, rather than a same-allele comparison; it does not validate either BALF5-TALDO1 pair or establish biological mimicry.

1.4.1 PANDORA/MODELLER register stress test

PANDORA/MODELLER was configured as a forced-template HLA-DRB1*15:01 register stress test. The primary template was 1BX2 and the sensitivity template was 6CQQ; these templates were preselected rather than discovered by an automated template search. For each template, the prepared file retained the first pMHC copy only, remapped its HLA-alpha, HLA-beta, and peptide chains to M/N/P, and excluded TCR and duplicate pMHC chains. The HLA target sequences were HLA-DRA*01:01 and HLA-DRB1*15:01. Template metadata specified peptide anchors 5, 8, 10, and 13 for 1BX2 and 3, 6, 8, and 11 for 6CQQ. A local PANDORA override retained target termini during alignment. Each template–peptide–register condition requested 20 models using slow loop refinement and at most four local jobs. The runtime was pinned to PANDORA commit `618bbb573645bd428697666475a2e47a2e418d01`; the local environment record includes a MODELLER/PANDORA import probe and explicit environment lock.

The reciprocal calibration comprised 120 scored models. The predeclared pass rule required, for each expected register, complete 20-model production, no target-template leakage, at least one top-five expected-register core-backbone RMSD below 2 Å, and an expected-register top-five median lower than both adjacent-register decoys. Both reciprocal controls passed. Technical ensemble members are not biological replicates, and PANDORA/MODELLER scores, model energies, contacts, or RMSD values are not binding affinities or evidence of TCR recognition.

Poisson-Boltzmann/APBS pMHC surface-potential comparison was explored as an exploratory surface-resemblance descriptor [12]. Its score was excluded from primary ranking because a 30-result control gate yielded 15 failed and 12 missing results. This did not test or refute biological T-cell cross-reactivity. An AlphaFold 3 TCR-pMHC calibration against 1YMM and 2WBJ was also assessed with DockQ and CAPRI pose-recovery criteria [13, 14]; all ten returned models scored below the DockQ acceptable-quality boundary of 0.23 and both controls were CAPRI “Incorrect,” so these outputs were excluded from candidate TCR ranking.

Separately, TCRModel2 generated five-component TCR-pMHC models comprising HLA-alpha, HLA-beta, peptide, TCR-alpha, and TCR-beta chains [15]. The calibration references 1YMM, 1ZGL, and 2WBJ each received five ranked models. DockQ and CAPRI classifications were evaluated alongside TCR-placement, pMHC, TCR-fold, and whole-complex RMSD. 1ZGL produced five acceptable models (DockQ 0.411-0.479); 1YMM produced five incorrect models (0.010-0.140); and 2WBJ produced five incorrect models (0.016-0.038). Only one control met the recovery criterion. Candidate outputs were excluded from ranking and cannot support shared-TCR or cross-reactivity claims; BALF5 replicate models remain exploratory internal predictions rather than docking validation.

Table S1: Prespecified method-gating summary. A method could influence candidate ranking only if its calibration gate was passed.

Method	Intended role	Calibration and gate	Observed result	Used for candidate ranking?
APBS surface potential	Exploratory pMHC surface resemblance	30 required control outputs; complete, comparable calculations required	15 failed; 12 missing; the three usable outputs did not discriminate the calibration control from decoys	No
AF3 TCR-pMHC	Exploratory TCR-pMHC pose recovery	1YMM and 2WBJ; DockQ $\geq$ 0.23 and non-incorrect CAPRI pose required	10 returned models; all below DockQ 0.23 and both controls CAPRI Incorrect	No
TCRModel2	Exploratory TCR-pMHC pose recovery	1YMM, 1ZGL, and 2WBJ; recovery required across the calibration set	15 returned models; 1ZGL acceptable (5/5), 1YMM and 2WBJ incorrect (0/5 each)	No

2 Supplementary Figures

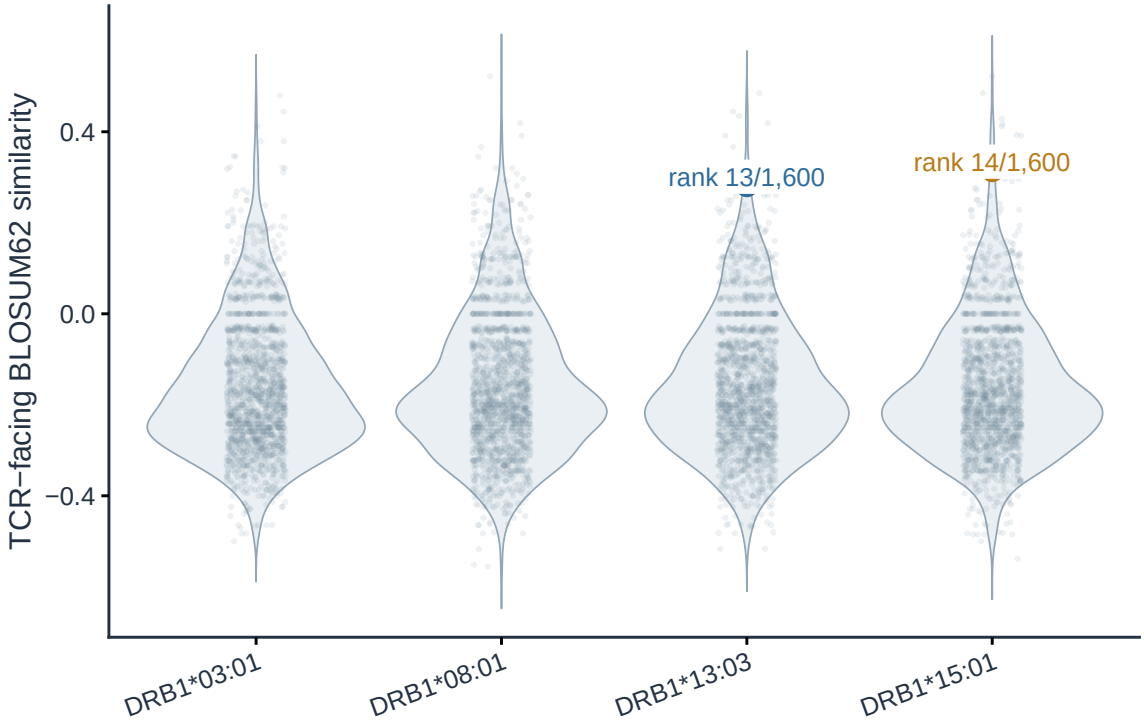


Figure S1: **Within-allele distributions of TCR-facing BLOSUM62 similarity scores.** Each point represents one EBV-human peptide comparison ($n = 1{,}600$ per HLA allele); violin profiles summarize the allele-specific score distributions. Labels identify the BALF5-TALDO1 pairs selected for evidence review. Reported ranks are calculated within each allele, not across the combined 6,400-pair dataset.

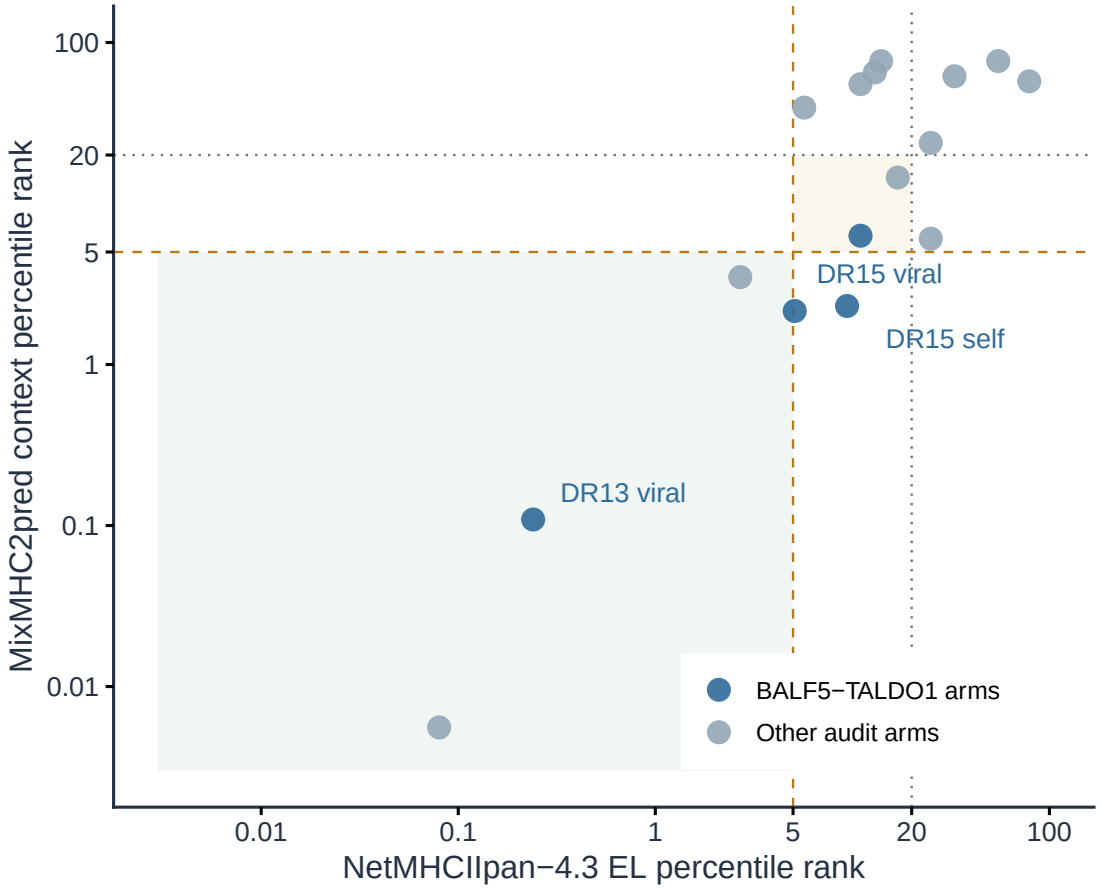


Figure S2: **Concordance of peptide-HLA class II binding-percentile predictions in the initial eight-pair audit.** Each point represents one peptide arm. NetMHCIIpan-4.3 EL and MixMHC2pred context percentile ranks are plotted on logarithmic axes; lower values indicate stronger predicted binding. Gold dashed lines mark the $\leq$ 5% full-consensus threshold, gray dotted lines mark the $\leq$ 20% lesser-support threshold, and the shaded region identifies arms below 5% in both predictors. Blue points are BALF5-TALDO1 arms; gray points are other audited arms.

3 Supplementary Tables

Table S2: Top sequence-ranked pairs within each allele.

Allele	Rank 1	Rank 2	Rank 3
*03:01	EBNA2$_{136\text{-}150}$-MOG$_{97\text{-}109}$ (0.48)	BNRF1$_{37\text{-}51}$-CLDN11$_{193\text{-}207}$ (0.44)	BALF4$_{103\text{-}116}$-CRYAB$_{1\text{-}15}$ (0.41)
08:01	EBNA1$_{87\text{-}101}$-ANO2$_{79\text{-}93}$ (0.52)	BLLF1$_{65\text{-}75}$-PLP1$_{140\text{-}152}$ (0.42)	BZLF1$_{198\text{-}210}$-MAG$_{612\text{-}626}$ (0.391)
13:03	EBNA3B$_{139\text{-}153}$-MOG$_{221\text{-}235}$ (0.49)	BNRF1$_{37\text{-}51}$-CLDN11$_{193\text{-}207}$ (0.44)	BARF1$_{1\text{-}15}$-MAG$_{1\text{-}15}$ (0.44)
15:01	EBNA1$_{87\text{-}101}$-ANO2$_{79\text{-}93}$ (0.52)	EBNA3B$_{139\text{-}153}$-MOG$_{221\text{-}235}$ (0.49)	BALF4$_{575\text{-}589}$-CNP$_{1\text{-}15}$ (0.43)

Table S3: Initial eight-candidate evidence audit.

Candidate pair HLA context	Predicted support V / S	Register V / S	IEDB arms	Ligand arms	Audit status
EBNA2-MOG					
HLA-DRB1*03:01	No / Yes	Yes / Yes	0	0	Hold
BARF1-CRYAB					
HLA-DRB1*03:01	No / No	No / Yes	0	0	Hold
EBNA1-ANO2					
HLA-DRB1*08:01	No / No	Yes / Yes	0	0	Hold
EBNA-LP-MOG					
HLA-DRB1*08:01	No / No	No / Yes	0	0	Hold
EBNA3B-MOG					
HLA-DRB1*13:03	No / No	No / No	0	0	Hold
BALF5-TALDO1					
HLA-DRB1*13:03	Yes / No	Yes / Yes	0	0	**Medium** priority
BALF4-CRYAB					
HLA-DRB1*15:01	Yes / No	Yes / Yes	1	1	Hold
BALF5-TALDO1					
HLA-DRB1*15:01	No / No	Yes / Yes	1	0	**Medium** priority

Note. V/S, viral/self peptide arm. Predicted presentation support: both NetMHCIIpan-4.3 EL and MixMHC2pred output ranks $\leq 5\%$ for that arm; it is not measured affinity. Register: both outputs selected the same declared P1-P9 core. IEDB and ligand entries are numbers of peptide arms with exact-HLA supporting records (0-2); they are not numbers of studies or patients. These fields do not establish natural presentation or T-cell cross-reactivity.

Table S4: Secondary provisional computational strata in the 49-pair cohort.

Target allele	Viral-human pair	Status	Supporting factor	Limitation
DRB1*15:01	$BALF5_{627\text{-}641}$-$TALDO1_{216\text{-}230}$	Stratum A	Three selected outputs met 20% routing rule; viral IEDB binding	Permissive threshold; register instability; no self-arm elution record
DRB1*13:03	$BALF5_{627\text{-}641}$-$TALDO1_{108\text{-}122}$	Stratum B	Three selected outputs met 20% routing rule	Permissive threshold; unresolved geometry; no verified public record
DRB1*15:01	$EBNA1_{87\text{-}101}$-$MBP_{217\text{-}233}$	Stratum B	Declared register retained across calibrated PANDORA templates	Binding-predictor discrepancy; lower rank
DRB1*15:01	$BALF4_{103\text{-}116}$-$CRYAB_{1\text{-}15}$	Stratum C	Initial sequence similarity	Template discordance and register sensitivity
DRB1*03:01	$EBNA2_{136\text{-}150}$-$MOG_{97\text{-}109}$	Stratum C	Initial sequence rank	No same-allele PANDORA calibration
DRB1*08:01	$BHRF1_{122\text{-}133}$-$ANO2_{104\text{-}118}$	Stratum C	Initial sequence rank	No same-allele PANDORA calibration
DRB1*08:01	$EBNA1_{87\text{-}101}$-$ANO2_{79\text{-}93}$	Stratum C	Initial sequence rank	No same-allele PANDORA calibration
Various	Various	Untiered	Heterogeneous profiles	Incomplete consensus or evidence

Table S5: Constrained-null simulation results for the eight audited pairs. Ten thousand replicates were completed independently for each allele at random seed 20260914. The 95% null interval reports the 2.5th and 97.5th percentiles.

Target	Allele	Pair	Observed score (rank)	Null median	95% null interval	Monte Carlo p	BH q	maxT p
HY03_SEQ_01	DRB1*03:01	EBNA2-MOG	0.48000 (1)	-0.18519	[-0.35714, 0.28000]	0.00790	0.06319	0.83572
HY03_SEQ_02	DRB1*03:01	BARF1-CRYAB	0.27273 (13)	-0.17949	[-0.32500, 0.34375]	0.08189	0.08189	1.00000
HY08_SEQ_01	DRB1*08:01	EBNA1-ANO2	0.52174 (1)	0.00000	[-0.32000, 0.52174]	0.03420	0.07199	0.66483
HY08_SEQ_02	DRB1*08:01	EBNA-LP-MOG	0.32000 (6)	-0.11538	[-0.46154, 0.32000]	0.06539	0.07634	1.00000
HY13_SEQ_01	DRB1*13:03	EBNA3B-MOG	0.48485 (1)	-0.13514	[-0.32432, 0.48485]	0.03600	0.07199	0.82102
HY13_SEQ_02	DRB1*13:03	BALF5-TALDO1	0.28125 (13)	-0.15625	[-0.33333, 0.25806]	0.01720	0.06879	1.00000
HY15_SEQ_01	DRB1*15:01	BALF4-CRYAB	0.41176 (5)	-0.10000	[-0.30233, 0.47059]	0.05559	0.07634	0.98260
HY15_SEQ_02	DRB1*15:01	BALF5-TALDO1	0.31429 (14)	0.05714	[-0.18919, 0.41176]	0.06679	0.07634	1.00000

Sequence-score calibration only; not evidence of endogenous presentation, pMHC stability, TCR recognition, cross-reactivity, molecular mimicry, or an MS mechanism.

Table S6: Lead-pair sequences, frozen cores, and continuous computational-output percentiles. Values are percentile ranks; lower is more favorable within each output. They are retained for audit, not combined as independent evidence or interpreted as measured affinity.

Allele	Arm	Full 15-mer	Declared core	IEDB rec.	NetMHCIIpan EL	MixMHC2pred context
DRB1*13:03	BALF5	TGGVYHFVKKHVHES	YHFVKKHVH	8.70	0.24	0.153
DRB1*13:03	TALDO1	DARLSFDKDAMVARA	LSFDKDAMV	11.0	11.0	6.31
DRB1*15:01	BALF5	TGGVYHFVKKHVHES	VYHFVKKHV	20.0	5.10	3.37
DRB1*15:01	TALDO1	SVTKIYNYYKKFSYK	IYNYYKKFS	7.00	9.40	2.31

Note. The strict all-three-output, both-arm routing rule fails for both pairs at 10% and below; both pass only at 20%. The IEDB-recommended output is an archival service dependency whose internal version and tie behavior were not retained.

Table S7: Declared P1–P9 cores for the two BALF5-TALDO1 sequence-prioritized comparisons. Bold residues are canonical positional anchors P1, P4, P6, and P9; unbolded residues are nominally TCR-exposed/core non-anchor positions P2, P3, P5, P7, and P8. This positional convention does not imply uniform anchor chemistry across alleles. These are frozen V3 input cores, not experimentally mapped registers.

Allele	Peptide arm	Declared core (P1–P9)
DRB1*13:03	$BALF5_{627\text{-}641}$	**Y**HF**V**K**K**HV**H**
DRB1*13:03	$TALDO1_{108\text{-}122}$	**L**SF**D**K**D**AM**V**
DRB1*15:01	$BALF5_{627\text{-}641}$	**V**YH**F**V**K**KH**V**
DRB1*15:01	$TALDO1_{216\text{-}230}$	**I**YN**Y**Y**K**KF**S**

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